Conditions and Loops

Course: Problem Solving

Learning Objectives:

- Understand conditions and loops

- Learn if, if-else, elif

- Understand for and while loops

- Apply logic using examples

What are Conditions?

A **condition** is a rule or expression that gives:

True

False

Purpose of conditions:

Used for decision making

Helps choose the correct action

Examples:

If it is raining → take umbrella

If marks ≥ 40 → pass

If light is red → stop

Why Conditions are Important

Conditions help to:

Make logical decisions

Handle different situations

Build intelligent systems

Real-world uses:

Exam result systems

ATM transaction approval

Login systems

Online shopping discounts

Without conditions, every program would behave the same way in all situations.

IF Statement

Definition of IF:
The **if statement** executes an action **only when the condition is true**.

Meaning in simple words:

One condition → one action

Examples:

If student studies → gets good marks

If battery is low → charge phone

If temperature is high → turn on fan

IF–ELSE Statement

Definition of IF–ELSE:
The if–else statement provides two choices:

If condition is true → do one action

Else (condition is false) → do another action

Examples:

If marks ≥ 40 → Pass

Else → Fail

Real life:

If internet is available → attend class

Else → inform teacher

ELIF Statement

Definition of ELIF:
The **elif statement** is used when there are **more than two conditions**.

Meaning in simple words:

Many conditions → different outcomes

Examples:

If marks ≥ 90 → Grade A

Else if marks ≥ 75 → Grade B

Else → Grade C

Real life:

If weather is hot → use AC

Else if weather is normal → use fan

Else → wear warm clothes

What are Loops?

A **loop** is a method of **repeating the same task multiple times**.

Why loops are used:

To avoid writing same work again and again

To save time

To handle large amount of data

Examples:

Teacher checking all answer sheets

Attendance of all students

Sending message to every student

Why Loops are Important

Without loops:

Work becomes slow

Repetition is manual

Programs become lengthy

With loops:

One logic handles many items

Faster execution

Better efficiency

Example:

Calculating marks of 5 students manually

Calculating marks of 500 students using loop

FOR Loop

A **for loop** is used when we **already know how many times** the task must repeat.

Simple idea:

Fixed number of repetitions

Examples:

Print numbers from 1 to 10

Take attendance of 60 students

Distribute 30 answer sheets

Check 50 records

WHILE Loop

A **while loop** repeats the task **as long as the condition remains true**.

Simple idea:

Repetition depends on condition

Examples:

Study while exam is not over

Keep entering password while it is wrong

Fill water bottle while it is not full

Wait while bus has not arrived

Conditions vs Loops

Conditions	Loops
Used for decision making	Used for repetition
Based on True/False	Based on repeated execution
Executes once	Executes many times
Examples: if, if–else, elif	Examples: for, while
Example: Pass/Fail	Example: Attendance of all students

Summary

- - If → one decision
- - If-Else → two decisions
- - Elif → many decisions
- - For → fixed repetition
- - While → condition-based repetition